EE 709 Assignment 1

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You are given a sequential circuit with inputs $reset, r_0, r_1$, state variables q_0, q_1, p and outputs g_0, g_1, described by the following equations:

$$\begin{aligned}
q_0(k+1) &= \overline{reset(k)}.(r_0(k).\overline{q_0(k)} \ + \ \overline{g_0(k)}.q_0(k)) \\
q_1(k+1) &= \overline{reset(k)}.(r_1(k).\overline{q_1(k)} \ + \ \overline{g_1(k)}.q_1(k)) \\
p(k+1) &= \overline{reset(k)}.(p(k) \ \oplus \ (g_0(k) + g_1(k))) \\
g_0(k) &= (p(k) \ ? \ q_0(k).\overline{q_1(k)} \ : \ q_0(k)) \\
g_1(k) &= (p(k) \ ? \ q_1(k) \ : \ q_1(k).\overline{q_0(k)})
\end{aligned}$$

1 Reachable set using BDD's

Determine, using ROBDD's. the set of reachable states (p, q_0, q_1) assuming that $reset(0) = 1$, and $reset(k) = 0$ for all $k > 0$. The other inputs are not constrained in any way. Enumerate the states that can be reached.

1.1 Hints

During this operation you will need to do a substitution of variables. In order to do this, use the following sequence of operations

```
// Given bdd f on variables x0,x1,x2.
//
// We want to find the bdd obtained by
// substituting y0 for x0, y1 for x1, y2 for x2.
//
// First create an association
```

```
bdd Q[4];
Q[0] = y0; Q[1] = y1, Q[2]=y2; Q[3] = 0;
int assoc = bdd_new_assoc(bddm,Q,0);

// set the association for further
// operation in the bdd manager.
bdd_assoc(bddm,assoc);

// substitute x's by y's
// bdd fy will be the result of
// the substitution
bdd fy = bdd_substitute (bddm,f);
```

2 Property checking using SAT

1. Determine, using SAT, if the following properties hold

   (a) $\mathbf{AG}(r_0 \Rightarrow Fg_0)$.

   (b) $\mathbf{AG}(r_1 \Rightarrow Fg_1)$.

   (c) $\neg\mathbf{E}F(g_0.g_1)$.

   Note: establish your proof over a window of eight cycles.

2. Now assume that the circuit is modified by making $p(k) = 0$ for all k. Determine, using SAT, if the property $\mathbf{AG}(r_1 \Rightarrow Fg_1)$ still holds.

   Note: establish your proof over a window of eight cycles.

3 Instructions

- Please submit a PDF document which summarizes your entire approach.
- For each step in your PDF document, please attach evidence that you have actually done what you have claimed to do. That is, attach code, run logs etc.

- 50% of the grade will be determined by the quality and completeness of your PDF document. Note: quality means clarity, correctness, completeness and most importantly consiseness.
- Submission is due in two weeks starting from 28 September.